

- Endoderm, **680**, **1049**, 1050*f*–1051*f*
 Endodermis, **768**, **792**, 793*f*
 Endomembrane system, **104**
 bound ribosomes and, 354*f*
 endoplasmic reticulum of, 104, 105*f*
 Golgi apparatus, 105, 106*f*, 107
 lysosomes, 107*f*, 108
 signal mechanism for targeting polypeptides to, 354*f*
 vacuoles, 108*f*
 Endometriosis, **1033**
 Endometrium, **1026**, 1027*f*
 Endomycorrhizae, **656**, 660, 663, **817**, 818*f*
 Endoparasites, **1220**
 Endophytes, **667f**, 668, **814**
 Endoplasmic reticulum (ER), **104**, 109*f*
 animal cell, 100*f*
 as biosynthetic factory, 104, 105*f*
 cellular membrane synthesis and, 131*f*
 ribosomes and, 103*f*, 104
 in RNA virus replicative cycle, 405*f*
 targeting polypeptides to, 354*f*, 355
 Endorphins, 40*f*, 78, **1081t**
 Endoskeletons, 1132*f*–1134*f*, **1133**
 Endosperm, **647**, 826, **826**, 827*f*–829*f*
 Endospores, **575f**
 Endosymbiont theory, **109**, 110*f*, 189, 534, 535*f*
 Endosymbiosis, 109, 110*f*, **534**, 535*f*, **594**–595, 596*f*–597*f*, 608*f*, 609, 617
 Endothelin, 931
 Endothelium, blood vessel, **929f**
 Endothermic organisms, **736**, **884f**, 885*f*–889*f*, 890–892
 Endotoxins, **588**
 Energetic hypothesis on food chain length, **1225f**
 Energy, **32**, **144**
 allocation and use of, 889, 890*f*
 animal. *See* Bioenergetics
 in arctic tundra ecosystem, 1256*f*–1257*f*
 ATP energy coupling and, 150, 151*f*–153*f*
 biofuel technology to reduce dependence on fossil fuels for, 838
 chemical, 144, 889, 890*f*
 chemiosmosis as energy-coupling mechanism, 175, 176*f*, 177
 conservation of, 1239
 in ecosystems, 1238*f*–1240*f*, 1241
 electrons and levels of, 32*f*, 33–34
 in energy flow and chemical cycling, 164*f*, 165
 enzymatic catalysis and. *See* Enzymatic catalysis
 forms of, 144, 145*f*
 free-energy change and. *See* Free-energy change
 global human use of, 1210*f*, 1211
 heat as, 46
 hydrothermal vents, prokaryotes, and, 592
 kinetic, 46, 144, 145*f*, 163
 laws of transformation of, 145*f*–146*f*, 147
 light. *See* Light energy
 locomotion and, 1135–1136
 metabolism and. *See* Metabolism
 in photosynthesis, 41*f*
 potential, 32, 144, 145*f*, 163
 primary production of, 1241, 1242*f*–1244*f*, 1243*t*, 1245
 processing of, as theme of biology, 2, 3*f*
 regulation of storage of, in animal nutrition, 915, 916*f*, 917
 secondary production of, 1246*f*–1247*f*, 1248
 storage of, in fats, 74
 thermal, 46, 132, 144
 transfer and transformation of, as theme of biology, 9*f*, 143
 transformation of, 109, 110*f*–111*f*, 209*f*
 trophic levels and, 1240*f*, 1241, 1246, 1247*f*, 1248
 Energy and matter
 bioenergetics, 163
 decomposition and, 1259
 eukaryotic cells and, 93*f*
 hibernation, 897
 photosynthesis and, 211
 reproductive success and, 1042
 as theme of biology, 2, 3
 transfer and transformation of, 9*f*
 Energy budgets
 animal, 892
 ecosystem, 1241, 1242*f*
 Energy conservation, animal, 892, 893*f*
 Energy coupling, **150**, 151*f*–153*f*
 Energy flow
 in ecosystems, 1238*f*–1240*f*, 1241, 1256*f*–1257*f*
 between trophic levels, 1246*f*–1247*f*, 1248
 Energy flow and chemical cycling, 9*f*, 164*f*, 165.
 See also Biogeochemical cycles; Primary production; Secondary production
 Energy pyramids, 1246, 1247*f*, 1248
 Engelmann, Theodor W., 193, 194*f*
 English ivy, 784*f*
 Enhancers, **374f**–375*f*, 397
 Entamoebas, 613
 Enteric division, peripheral nervous system, 915, **1088f**–1089*f*
 Enthalpy, 147
 Entomopathogenic fungus, 672
Entomophthora muscae, 662*f*
 Entropy, 143, **146**–147, 1239
 Entry stage, phage lytic cycle, 402*f*
 Enveloped viruses, 401, 405*f*
 Environment
 adaptation to, 2*f*
 adaptive evolution as fitness to, through natural selection, 494, 498, 499*f*. *See also* Adaptive evolution
 animal and plant responses to, 894*f*
 behavior and stimuli from, 1140*f*–1141*f*
 biotechnology in cleanup of, 418*f*, 437
 bottleneck effect from changes in, 495*f*, 496
 cancer and factors of, 394–395
 C. Darwin on natural selection and adaptations to, 14*f*–16*f*. *See also* Adaptations; Evolution; Natural selection
 cell cycle control system and factors of, 247, 248*f*
 cellular membranes and factors of, 128*f*, 129
 Earth's early, and origin of life, 526*f*–528*f*
 ecology as interactions between organisms and. *See* Ecology
 enzymatic catalysis and factors of, 157, 158*f*, 159
 evolution and, 485
 exchanges with, in animals, 874, 875*f*, 876
 as factor in red maple leaf structure, 762
 gene regulation in response to, 377
 heat exchange with, in animals, 873*f*, 885*f*–889*f*
 hormones and, 1018
 human impacts on. *See* Human environmental impacts
 hybrid zones and, 517*f*
 impact of, on phenotypes, 282*f*
 induction in differential gene expression as response to, 382*f*, 383
 influence of evolution and, on nitrogenous wastes, 983
 interaction with, as theme of biology, 10*f*–11*f*
 metagenomics and genome sequencing of groups of species in, 444
 nervous system disorders and, 1102*f*
 organism interactions with, 1165, 1178, 1183*f*–1186*f*
 as organisms and their surroundings, 469
 plant responses to abiotic stresses from, 797–798, 862–863, 864*f*, 865
 population ecology as study of populations in, 1190*f*, 1191. *See also* Population ecology
 population growth in unlimited, 1196, 1197*f*
 prokaryotic adaptations to extreme conditions in, 573*f*, 574
 protein structure and factors of, 82*f*, 83
 reproductive cycles and cues from, 1021
 response to, as property of life, 3*f*
 strength of ionic bonds and factors in, 38
 vertebrate kidney adaptations to diverse, 991*f*–993*f*
 Environmental issues. *See also* Climate change; Ecology; Pollution
 decline in amphibian populations, 733
 density-dependent population regulation by toxic waste, 1204*f*
 prokaryotes, biotechnology, and, 589, 590*f*–591*f*
 threats to biodiversity, 1260*f*, 1261, 1263, 1264*f*–1266*f*
 threats to seed plant diversity, 651*f*, 652
 Environmental toxins, 1275, 1276*f*–1277*f*
 Enzymatic catalysis. *See also* Enzymes
 activation energy barrier and, 153, 154*f*
 allosteric regulation of, 160*f*–161*f*
 of cellular respiration, 183*f*, 184
 cofactors and, 158
 effects of temperature and pH on, 157, 158*f*
 in enzyme active sites, 156*f*
 enzyme inhibitors and, 158, 159*f*
 evolution of enzymes and, 159*f*
 graphing, for blood glucose levels, 157
 lowering of activation energy by enzymes in, 154, 155*f*
 in plant cells, 208*f*
 regulation of, 366*f*–369*f*, 370
 by ribozymes, 346
 substrate specificity of enzymes in, 155*f*, 156
 Enzymatic hydrolysis, 904, 908*f*
 neurotransmission and, 1079, 1080*f*
 in vacuoles, 108
 Enzymatic proteins, 76*f*
 Enzymes, **67**, **153**. *See also* Enzymatic catalysis
 active sites of, 155*f*–156*f*
 activity measurement of, 163
 as catalysts, 153
 in cell-signaling nuclear responses, 226, 227*f*
 in chemical digestion, 907*f*–908*f*, 909
 evolution of, 159*f*
 facilitation of synthesis and breakdown of polymers by, 67*f*
 fungal, 654–655
 in gastric juice, 907*f*–908*f*
 gene relationship with, in protein synthesis, 336*f*–338*f*
 inducible and repressible, 367*f*–368*f*, 369
 locations of, in cells, 161*f*
 lowering plasma LDL levels by inactivating liver, 938
 lysosomes and, 107*f*, 108
 as proteins, 75, 76*f*, 130*f*
 restriction. *See* Restriction enzymes
 ribozymes as, 346
 in saliva, 906
 smooth ER and rough ER, 104–105
 substrate specificity of, 155*f*, 156
 from *Thermus aquaticus*, 441
 Enzyme-substrate complexes, **155f**, 156
 Eosinophils, 934*f*–935*f*, **955**
 Ependymal cells, 1090*f*
Ephedra, 642*f*
 Ephrussi, Boris, 336
 Epiblast, 1052*f*–1053*f*
 Epicotyl, **829f**–830*f*
 Epidemics, **409f**, 410, 867
 Epidemiology, 902
 Epidermis, 5*f*, **763f**
 Epididymis, **1025f**
 Epigenetic inheritance, **372f**, 373
 Epigenetic modifications, cancer gene, 388, 389*f*
 Epigenetics, 430
 Epiglottis, 906*f*
 Epilepsy, 1076, 1098
 Epinephrine (adrenaline), **1002**, 1003*f*
 adrenal glands and, 1004*f*, 1012*f*–1013*f*
 in fight-or-flight responses, 212*f*, 928
 multiple effects of, 1004*f*, 1013*f*
 second messenger of, 223*f*–224*f*
 signal transduction pathway of, 216, 226, 227*f*, 233
 as water-soluble hormone, 1001*f*
 Epiphytes, 632*f*, 818, **819f**, 820
 Epistasis, **281f**
 Epithalamus, 1093*f*
 Epithelial milk-secreting cell, 392*f*
 Epithelial tissue, **877f**
 as barrier defense, 954
 cell junctions in, 120*f*
 in small intestine, 909*f*
 transport, 981, 982*f*, 988*f*, 989
 Epitopes, **958**, 959*f*, 969, 976
 Epstein-Barr virus, 394
 Equilibrium
 chemical. *See* Chemical equilibrium
 mechanoreceptors for sensing, 1112*f*–1116*f*
 population, 1203*f*
 Equilibrium potential (Eion), **1071f**
 Equine encephalitis virus, 401
Equisetum, 632*f*, 633
Equus, 548, 549*f*
 ER. *See* Endoplasmic reticulum
 Erectile dysfunction, 1026, 1081–1082
 Ergots, 669*f*